

Theoretical Exercise 5

1) We prove that minimizing $\sum_{j=1}^K \text{WCSS}(C_j)$ is equivalent to minimizing $\sum_{j=1}^K \widetilde{\text{WCSS}}(C_j)$:

$$\begin{aligned} \|x_a - x_b\|^2 &= (x_a - x_b)^T (x_a - x_b) = (x_a - \mu_j + \mu_j - x_b)^T (x_a - \mu_j + \mu_j - x_b) = \\ &= ((x_a - \mu_j)^T + (\mu_j - x_b)^T) ((x_a - \mu_j) + (\mu_j - x_b)) = (x_a - \mu_j)^T (x_a - \mu_j) + \\ &+ (x_a - \mu_j)^T (\mu_j - x_b) + (\mu_j - x_b)^T (x_a - \mu_j) + (\mu_j - x_b)^T (\mu_j - x_b) = \\ &= \|x_a - \mu_j\|^2 + 2(\mu_j - x_b)^T (x_a - \mu_j) + \|\mu_j - x_b\|^2 \end{aligned}$$

Now let's substitute what we got into $\widetilde{\text{WCSS}}$:

$$\begin{aligned} \widetilde{\text{WCSS}}(C_j) &= \frac{1}{|C_j|} \sum_{a,b \in C_j} \|x_a - x_b\|^2 = \\ &= \frac{1}{|C_j|} \cdot \sum_{a,b \in C_j} (\|x_a - \mu_j\|^2 + 2(\mu_j - x_b)^T (x_a - \mu_j) + \|\mu_j - x_b\|^2) = \\ &= \frac{1}{|C_j|} \left(\sum_{a,b \in C_j} \|x_a - \mu_j\|^2 + \sum_{a,b \in C_j} \|\mu_j - x_b\|^2 + 2 \sum_{a,b \in C_j} (\mu_j - x_b)^T (x_a - \mu_j) \right) = \\ &= \frac{1}{|C_j|} \sum_{a,b \in C_j} \|x_a - \mu_j\|^2 + \frac{1}{|C_j|} \sum_{a,b \in C_j} \|\mu_j - x_b\|^2 + \frac{1}{|C_j|} \sum_{a,b \in C_j} 2(\mu_j - x_b)^T (x_a - \mu_j) = \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{|C_j|} \cdot |C_j| \cdot \sum_{a \in C_j} \|x_a - \mu_j\|^2 + \frac{1}{|C_j|} \cdot |C_j| \cdot \sum_{b \in C_j} \|\mu_j - x_b\|^2 \\
&+ \frac{1}{|C_j|} \cdot |C_j| \cdot 2 \cdot \frac{1}{|C_j|} \cdot \sum_{a, b \in C_j} \underbrace{(\mu_j - x_b)^T}_{\mathbf{0}} \underbrace{(x_a - \mu_j)}_{\mathbf{0}} = \quad (\text{we saw in the lecture})
\end{aligned}$$

$$= \sum_{a \in C_j} \|x_a - \mu_j\|^2 + \sum_{b \in C_j} \|\mu_j - x_b\|^2 + 2 \cdot \frac{1}{|C_j|} \sum_{b \in C_j} \mathbf{0}^T \cdot \mathbf{0} =$$

$$= \sum_{a \in C_j} \|x_a - \mu_j\|^2 + \sum_{b \in C_j} \|\mu_j - x_b\|^2 = \text{WCSS}(C_j) + \text{WCSS}(C_j) =$$

$$= 2 \text{WCSS}(C_j)$$

$\uparrow$
 definition of WCSS for $i = a, b$

2) We will take the gradient of $J(u; D, C)$ with respect to u and set it to the zero vector:

$$J(u; D, C) = \sum_{i \in C} \|x_i - u\|^2 = \sum_{i \in C} \sum_{k=1}^d (x_{i_k} - u_{i_k})^T (x_{i_k} - u_{i_k})$$

$x_i = (x_{i_1}, \dots, x_{i_d})$
 $u = (u_{i_1}, \dots, u_{i_d})$
 $\downarrow$

$$\nabla J(u; D, C) = \frac{\partial}{\partial u_j} \left(\sum_{i \in C} \sum_{k=1}^d (x_{i_k} - u_{i_k})^T (x_{i_k} - u_{i_k}) \right) =$$

Partial derivative is not 0 only when $k=j$

$$= \sum_{i \in C} \sum_{k=1}^d \frac{\partial}{\partial u_j} (x_{i_k} - u_{i_k})^T (x_{i_k} - u_{i_k}) = \sum_{i \in C} 2(x_{i_j} - u_j)(-1) =$$

$$= -2 \sum_{i \in C} x_{i_j} - u_j$$

We will now set this to 0:

$$-2 \sum_{i \in C} x_{ij} - u_j = 0 \quad | : (-2)$$

u_j not dependant on i

$$\sum_{i \in C} x_{ij} - u_j = 0 \Rightarrow \sum_{i \in C} x_{ij} - \sum_{i \in C} u_j = \sum_{i \in C} x_{ij} - |C| u_j = 0$$

$$\sum_{i \in C} x_{ij} = |C| u_j \Rightarrow u_j = \frac{1}{|C|} \sum_{i \in C} x_{ij}$$

$$u = \frac{1}{|C|} \sum_{i \in C} x_i$$

3)

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Iteration 1:
Centroid 1: -0.981, 0.546, -0.671
Centroid 2: 1.252, 0.144, 0.756
Iteration 2:
Centroid 1: -0.831, 0.475, -0.613
Centroid 2: 1.381, 0.165, 0.875
Iteration 3:
Centroid 1: -0.831, 0.475, -0.613
Centroid 2: 1.381, 0.165, 0.875
Converged after 3 iterations.
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Same as the before, convergence check

* The standard k-means yields the same assignments to clusters. We will see it here:

Weighted:

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Iteration 1:
Centroid 1: -0.981, 0.546, -0.671
Centroid 2: 1.252, 0.144, 0.756
Cluster 0 points: [0 2 5 6]
Cluster 1 points: [1 3 4 7 8 9]
Iteration 2:
Centroid 1: -0.831, 0.475, -0.613
Centroid 2: 1.381, 0.165, 0.875
Cluster 0 points: [0 2 5 6 8]
Cluster 1 points: [1 3 4 7 9]
Iteration 3:
Centroid 1: -0.831, 0.475, -0.613
Centroid 2: 1.381, 0.165, 0.875
Cluster 0 points: [0 2 5 6 8]
Cluster 1 points: [1 3 4 7 9]
Converged after 3 iterations.
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Standard:

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Iteration 1:
Centroid 1: -0.860, 0.108, -0.550
Centroid 2: 1.223, 0.310, 0.650
Cluster 0 points: [0 2 5 6]
Cluster 1 points: [1 3 4 7 8 9]
Iteration 2:
Centroid 1: -0.644, 0.082, -0.480
Centroid 2: 1.424, 0.376, 0.820
Cluster 0 points: [0 2 5 6 8]
Cluster 1 points: [1 3 4 7 9]
Iteration 3:
Centroid 1: -0.644, 0.082, -0.480
Centroid 2: 1.424, 0.376, 0.820
Cluster 0 points: [0 2 5 6 8]
Cluster 1 points: [1 3 4 7 9]
Converged after 3 iterations.
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* Yes, the weighted-k-means algorithm always converges.

Let us first define the optimization problem:

$$I = \sum_{k=1}^K \sum_{x_i \in C_k} w_i \|x_i - \mu_k\|^2$$

clusters samples weight w_i distance

Assuming we have m samples, we can rewrite it as:

$$I = \sum_{j=1}^m w_j \|x_j - \mu_{c(x_j)}\|^2$$

where $\mu_{c(x_j)}$ is the mean of the cluster that x_j belongs to.

We will prove the convergence by using a reduction:

Let each data point x_i have an associated weight w_i , and let's construct a new dataset by duplicating each point x_i exactly w_i times:
(we assumed $w_i > 0$ for every i)

$$X' = \bigcup_{i=1}^n \{x_i, x_i, \dots, x_i\}$$

For any assignment of points to clusters, the total inertia on the above set is equal to the weighted-k-means inertia on the original set:

$$\sum_{i=1}^n w_i \|x_i - \mu_{c_i}\|^2 \iff \sum_{i=1}^N \|x'_i - \mu_{c'_i}\|^2$$

That's why the algorithm works exactly like it did before, and under this transformation, it is equivalent to the standard k -means algorithm convergence we saw in the lecture.